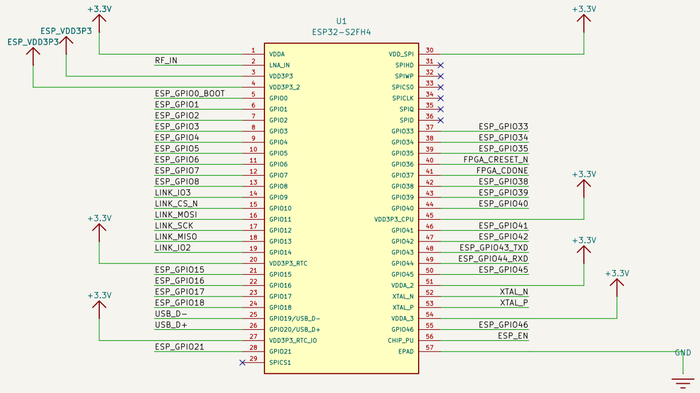
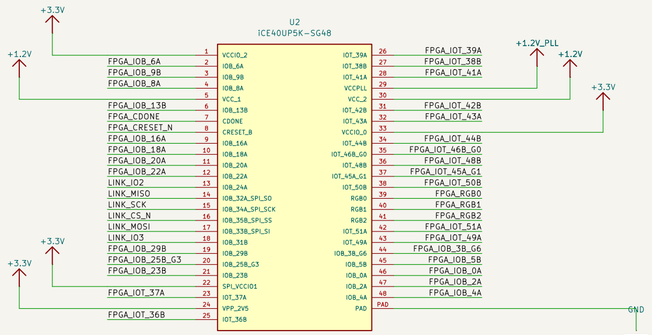
ESP32-S2 + iCE40UP5K || Development module | Rev A

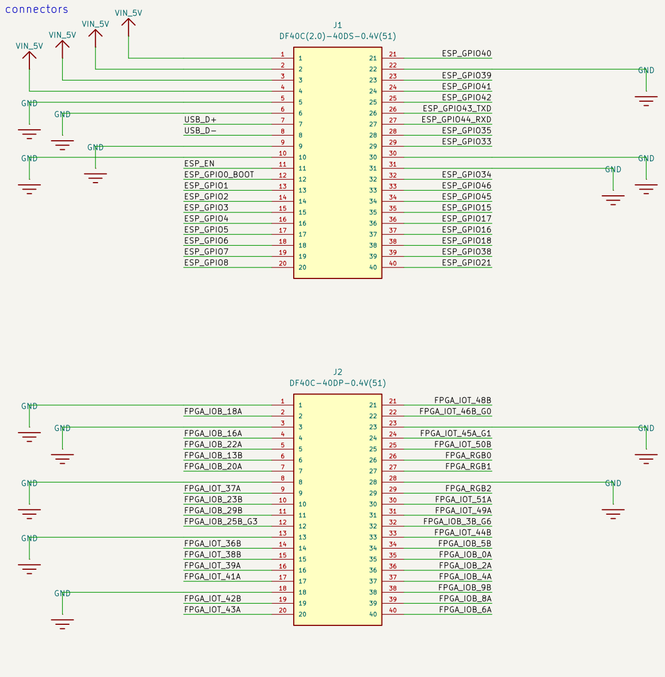
ESP32 / integrated flash



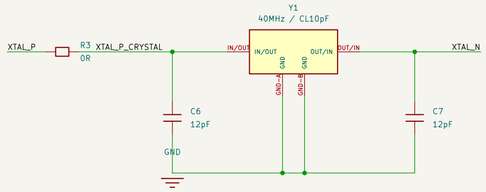
FPGA / shared processor interface



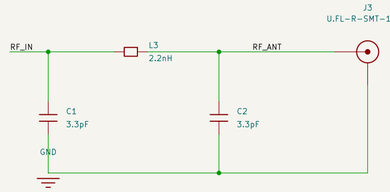
Motherboard connectors



40 MHz crystal

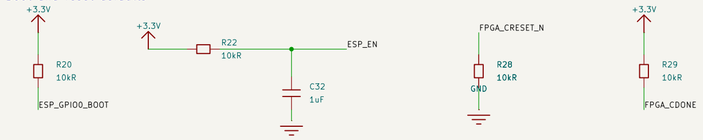


RF matching / external antenna

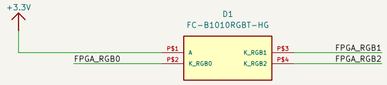


U.FL external antenna required for Wi-Fi.

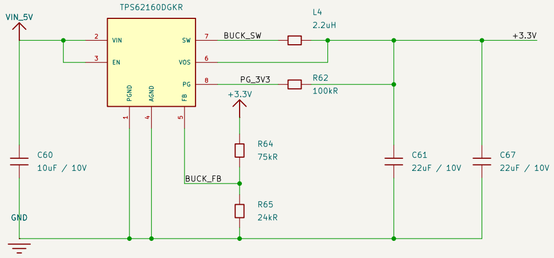
Boot and reset defaults



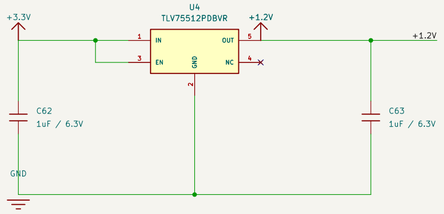
RGB status indicator



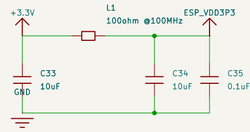
3.3 V buck regulator / 5 V input



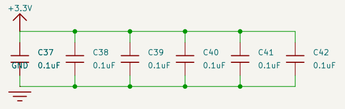
1.2 V LDO / FPGA core supply



ESP32 supply filtering / decoupling

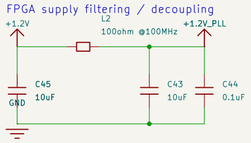


ESP32 pi filter: U1 pins 3, 4.

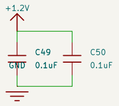


ESP32 3.3 V decoupling

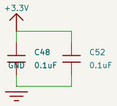
FPGA supply filtering / decoupling



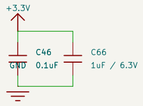
PLL pi filter: U2 pin 29.



Core: U2 pins 5, 30



Fixed 3.3 V supplies



Bank 0: U2 pin 33

FPGA debug test points





Sheet: /
File: module.kicad_sch

Title: ESP32 FPGA module – development

Size: A2 | Date: | **Rev: A-draft**

KiCad E.D.A. 10.0.6 | Id: 1/1